

Solenoid Active Damping

Open Source Presentation for RoboCup 2026

Two short, horizontal, olive-green bars are positioned on either side of the text, acting as decorative elements.

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The Pass Receiving Problem

Challenge:

- Receiving high velocity (3-5m/s) passes are hard to control
 - Ball bounces off before it can be controlled

Standard Solution:

- Complex mechanical systems (dribbler and compliant mechanisms)
 - Costly and time intensive to build, a steep barrier for new teams

Our Objective:

- Achieve the same damping performance as high-performing teams, without adding any parts

Energy going Unused?

Kicker ("Chicker")
Board

Example:

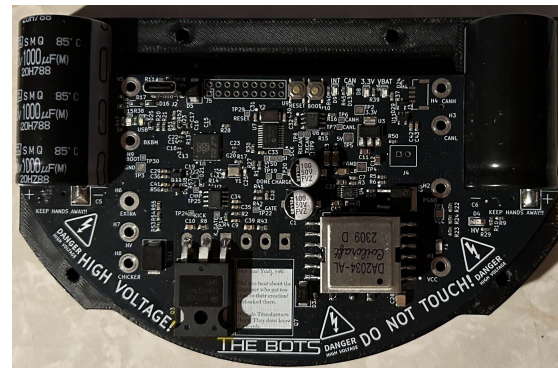
- Our hardware contains 2000uF caps charged to 210V
 - This is a total of 44.1J of energy!

Opportunity:

- Energy stored is dedicated to kicking
 - But, is unspent during pass receiving phase!

Question:

- Can we use the same energy to absorb the ball's kinetic energy?



Human-Centered Design

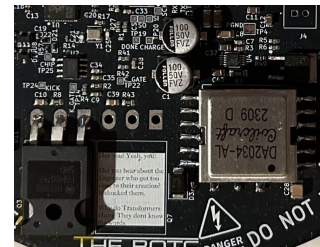


Human Analogy:

- A soccer player retracts their leg towards them as the ball gets to their feet, to stop the ball

Robot Adaptation:

- Extend and hold the kicker with minimal force, such that when the ball makes contact, the kicker collapses - transferring the energy
 - Uses existing standard kicker hardware used by most SSL teams



Firmware Implementation

Extension:

- Send a short pulse to produce a low power Kick

Wait:

- Allow time to pass for mechanical actuation of the kicker

Hold:

- Send a PWM signal to the IGBT gate drive
 - Duty cycle that equals the threshold that will keep the kicker extended

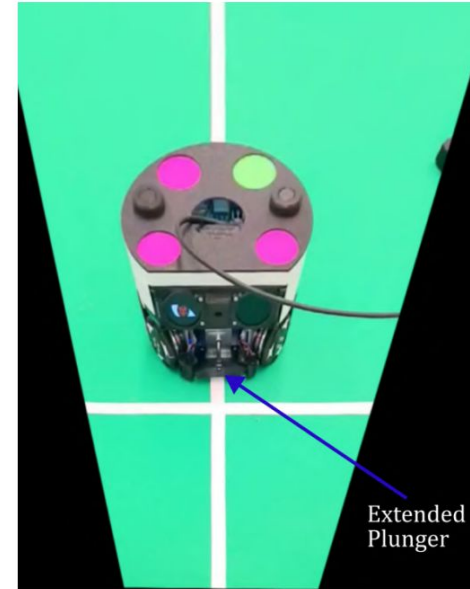
Early Testing (TDP Content)

Initial Results:

- Tested at 3 to 5 m/s of pre-collision velocity
 - Inconsistent outcomes with the same velocity

Our Observation:

- On a failed test, the plunger would re-extend after impact



Why Does it Work?

- Elastic Collision vs Inelastic collision
- We achieve a partially inelastic collision - to give way and absorb energy
 - With a perfect damping, we achieve a coefficient of restitution (e) of:

$$e = \frac{m_{ball}}{m_{plunger}} = \frac{0.045}{0.060} = 0.75$$

$$e = \frac{v'_{plunger} - v'_{ball}}{v_{ball}}$$

- Otherwise, e can be calculated using the momentum transfer and velocity difference
- Test 1: 3.0 m/s → e = 0.902 (bounced) | Test 2: 2.7 m/s → e = 0.75 (damped)

Recent Results

What we found:

- High Voltage Caps discharging at around 30V/s during hold

Solution:

- Added a duty cycle ramp to maintain the hold force

Result:

- Ball stops dead at the mouth of the robot or bouncing <1 robot diameter away

Future Work and Observations

- Damping performance will vary with plunger weight
- Static Friction modelling, before the plunger releases its hold
 - Momentum math assumes instantaneous collisions
 - Should be time dependent, impulse based for a more accurate results
 - Especially with a heavier plunger, as there will be more force to overcome

Thank You

Questions?

